Akash Kulkarni

Smyrna, Georgia (open to relocate) | 631-974-6592 | akashkulkarni530@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Full-Stack Engineer and Delivery Manager with 3.5+ years of experience building scalable applications using React, TypeScript, and AWS; architected a voice assistant deployed in 5 million vehicles, delivering features 75% faster; spearheaded the development of automated testing suites, reducing production defects by 50%; integrated GitLab CI/CD pipelines, improving release efficiency by 20%.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, Node.js, Java, C#, Asp.net, C++, OCaml.
Web, Mobile & AI: React, React Native, LLMs, Spring Boot, Express.js, Tailwind CSS, RESTful APIs, GraphQL.
Database & Cloud: AWS (EC2, S3, Serverless, RDS), PostgreSQL, MongoDB, Prisma ORM, Docker, CI/CD.
Testing & Tools: Cypress, JUnit, Mockito, Git, Unix, Bash, Unity, Unreal Engine, Pandas, Power BI.

WORK EXPERIENCE

Simulator & Virtual Reality Engineer, Intern July 2026 – Present
Pulseworks LLC

- Designed and deployed a scalable CMMS in React/TypeScript on AWS, managing 700+ assets and reducing scheduling overlaps by 40%.
- Resolved simulator video quality issues by executing daily asset maintenance and advanced hardware operations, enhancing simulation platform reliability by 25% for critical user applications.
- Reduced guest-facing mechanical faults by 25% by executing 50+ weekly simulation cycles and validating hardware synchronization, alongside ongoing troubleshooting of audio and networking issues.

Simulator & Virtual Reality Support Engineer, Intern May 2025 – May 2026
Pulseworks LLC

- Performed routine calibration on immersive VR hardware, increasing calibration accuracy by 15% via rigorous diagnostic testing.
- Created comprehensive digital inventories for 800+ VR simulation assets, enabling faster simulation setup by 30% and improving resource allocation.
- Investigated hardware performance logs to identify and address 50+ potential failures, preventing synchronization issues across Gforce platforms.

Software Engineer Associate | Delivery Manager January 2022 – July 2024
Cerence Inc.

- Promoted to Delivery Manager after demonstrating exceptional software engineering capabilities, expanding leadership remit to oversee the full SDLC for multiple automotive clients and ensure the timely delivery of complex voice modules.
- Architected a full-stack voice assistant for PSA Group in collaboration with Harman International, streamlining order entry processes and deploying code across a 5-million-vehicle fleet, addressing performance issues resulting in 75% faster feature delivery.
- Reduced debugging time by 30% by implementing modular code structures and reusable **TypeScript** and **JavaScript** component libraries, using multi-threaded and object-oriented development principles, and adhering to structural programming, coding standards, code reviews, TDD, and BDD.
- Orchestrated SDLC for Toyota, Subaru, and PSA Group voice modules, integrating GitLab CI/CD pipelines with a version control system, scripting deployment processes, and accelerating release velocity by 20%, enhancing application and QA processes.
- Spearheaded the creation of self-executing test suites using JUnit and Mockito for unit testing and relational database validation, reducing production defects by 50% while upholding stringent automotive safety standards.

Software Engineering, Intern May 2021 – August 2021
GKTCS Innovation

- Integrated asynchronous message queues and led UI A/B testing, translating insights into product improvements that increased user retention by 42%.
- Developed predictive models identifying user trends, leading to targeted content creation that increased organic site traffic by 80% through SEO strategies.

TECHNICAL PROJECTS

LLM-Powered Multi-Agency Transit Assistant | *Prompt Engineering*.

- Developed an LLM-powered transit trip planner using AI-assisted development tools like the Claude API, synthesizing MTA, PATH, and NJ Transit data into context-aware itineraries, and creating reporting tools for issues reported by 50+ users.
- Optimized system performance by implementing Chain-of-Thought, RAG, and Tool Use prompting strategies, ensuring 100% route correctness and maintaining departure-time accuracy within a 2-minute tolerance.

FindMySquad - Sports, Fitness, and Gaming Teammate Finder | *Web Programming*.

- Engineered a real-time coordination engine for a distributed application using Socket.IO and Node.js to support 200+ concurrent users, featuring a persistent messaging system and programmatic email notifications with sub-300ms backend latency.
- Implemented a comprehensive tournament management system with advanced validation for team registrations and scheduling, integrating Leaflet.js for geospatial discovery and preference-based matchmaking.

EDUCATION

Stevens Institute of Technology
Master of Science in Computer Science August 2024 – May 2026

MIT World Peace University
Master of Computer Applications July 2020 – June 2022

Mangalvedhekar Institute of Technology
Bachelor of Computer Applications August 2017 – April 2020